# João Pedro Paiva Pires

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I studied Physics, moved into data and machine learning, and I'm now in the final year of an MSc in Data Science and Engineering at IST. I work mainly in Python and SQL, building pipelines and models. I co-authored a paper under review at Nature Machine Intelligence and work on fraud detection with Feedzai Research. Available full time immediately.

## Education

**Instituto Superior Técnico**, MSc in Data Science and Engineering — Sept 2025 – July 2027

- **Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Information Processing and Retrieval, Computational Statistics, Data Analysis and Integration (Data Warehousing & ETL), Information Systems and Databases.

**University of Aveiro**, BSc in Physics — Sept 2022 – July 2025

- **Coursework:** Calculus, Linear Algebra, Discrete Mathematics, Numerical Methods, Statistical Physics.
- **Research Grant Award** – Recognized with a research grant during my final year, which I declined (exclusivity conflict).

## Experience

**Feedzai — MSc Research Project, Uncertainty-Aware Fraud Detection** — Sept 2026 – Present

- Building a Python pipeline over 24M+ card transactions, reshaping raw data into per-customer event streams.
- Modelling next-event distributions (amount, time) and scoring anomalies by predictive uncertainty.
- Designing experiments to benchmark these scores against supervised baselines, with a Feedzai Research Scientist.

**Caixa Geral de Depósitos — Summer Internship** — July 2026 – Aug 2026

*Ponta Delgada, São Miguel Island, Azores - Portugal*

- Advised clients on term deposits and investment funds, explaining risk-return trade-offs to a non-technical audience.
- Handled day-to-day client operations across banking products in a regulated environment, under data protection rules.

## Publications

**"Beyond the Surface: Estimating Internal Battery Pressure and Temperature with Optical Fiber Sensors and Machine Learning", Nature Machine Intelligence** — *Under-Review*

***L. Matuck, M. Nascimento**, J. P. Pires, J. F. F. Mendes, M. S. Ferreira*

- Built preprocessing pipelines for non-invasive optical fiber sensor signals and tuned TCN architectures via Bayesian optimization (500 trials), reaching $R^2$ of 0.996 (temperature) and 0.97 (pressure).

## Projects

**Data Warehouse & ETL Pipelines — Portugal: A Century of Transformation** — MySQL, Apache Hop

- Designed a star-schema data warehouse in MySQL (3 fact tables for economy, society and sector; shared country, time and sector dimensions) integrating Maddison Project, World Bank and historical-events data (1900–2025).
- Built staging and ETL pipelines in Apache Hop, with deduplication and value standardisation as data quality steps.
- Published an OLAP cube (Pentaho Schema Workbench) for analysis in Saiku and reports in Pentaho Report Designer.

**Personal Finance Tracker** — github.com/joaoopiress

- Text classification of transactions (Naive Bayes → Logistic Regression) with NLP-based category suggestions.
- R and PostgreSQL pipeline with automated ingestion, a 3-model spending forecast and a live Shiny dashboard.

**Deep Learning Architectures for Predicting PSI20 Trends** — github.com/joaoopiress

- Framed index movement prediction as a 3-class problem (up/flat/down) and benchmarked LSTM, Attention-LSTM and Transformer classifiers on 10 years of daily price and volume data (Macro F1 62.5%).
- Wavelet denoising and Z-score normalization; Bayesian hyperparameter search. **Best Project (18/20)**.

## Technical Skills & Interests

**Programming & Data:** Python (pandas, NumPy, scikit-learn, PyTorch), SQL (PostgreSQL, MySQL), R, Git/GitHub.

**ML & NLP:** classification, regression, time series, model evaluation, TF-IDF, Hugging Face, Matplotlib, Plotly.

**Cloud & LLMs:** AWS (preparing the ML Engineer – Associate certification), LLM prompting (Claude).

**Languages:** English (fluent), Portuguese (native). **Interests:** LLM Agents, Applied NLP, Data Products.